

Discrete Mathematics and Graph Theory

Module 5 (CO5) — Algebraic Structures and Applications

Questions with Detailed Step-by-Step Solutions

Every question is solved from zero to the final answer — what is used, why it is used, and every small step explained. Part A questions carry 2 marks; Part B questions carry 7 marks.

Part A — Short Questions (2 Marks)

Q1. (2 marks) Define a binary operation. Give an example. **Solution — full explanation.**

Definition. A **binary operation** on a set S is a function $f : S \times S \rightarrow S$; we write the result on (a, b) as $a * b$. The result must lie in S (closure). **Example.** Ordinary addition $+$ on $\mathbb{Z}$ is a binary operation (the sum of two integers is again an integer). Bitwise XOR $\oplus$ on $B = \{0, 1\}$ is another.

Final answer. A binary operation combines two elements of a set to give an element of the same set.

Q2. (2 marks) Define a semi-group and a monoid. **Solution — full explanation.**

Definitions.

- **Semi-group:** a set with an **associative** binary operation.
- **Monoid:** a semi-group that also has an **identity element**.

Final answer. Semi-group = associative operation; monoid = semi-group + identity.

Q3. (2 marks) Define a group. **Solution — full explanation.**

Definition. A **group** $(G, *)$ is a set G with a binary operation $*$ such that:

- **Closure:** $a * b \in G$ for all $a, b \in G$;
- **Associativity:** $(a * b) * c = a * (b * c)$;
- **Identity:** there is $e \in G$ with $a * e = e * a = a$;
- **Inverse:** for each a there is a^{-1} with $a * a^{-1} = a^{-1} * a = e$.

Final answer. A group is a set with an associative operation, an identity and inverses.

Q4. (2 marks) What is an abelian group? Give an example. **Solution — full explanation.**

Definition. A group $(G, *)$ is **abelian** (commutative) if $a * b = b * a$ for all $a, b \in G$. **Example.** $(\mathbb{Z}, +)$ is abelian (since $a + b = b + a$). Also $(\mathbb{Z}_n, +_n)$ and $\{1, -1, i, -i\}$ under multiplication.

Final answer. An abelian group is a group whose operation is commutative.

Q5. (2 marks) Define the order of a group and the order of an element. **Solution — full explanation.**

Definitions.

- **Order of a group** $|G|$: the number of elements in G .
- **Order of an element** a : the smallest positive integer n with $a^n = e$.

Final answer. $|G|$ = size of the group; order of a = smallest n with $a^n = e$.

Q6. (2 marks) Define a subgroup and state the subgroup test. **Solution — full explanation.**

Definition. A non-empty subset H of a group $(G, *)$ is a **subgroup** if H is itself a group under $*$. **Subgroup test (one-step).** $H \subseteq G$ is a subgroup if and only if for all $a, b \in H$, $a * b^{-1} \in H$.

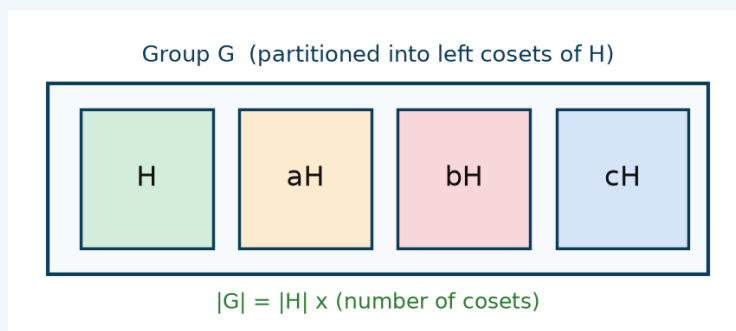
Final answer. H is a subgroup iff it is closed under $a * b^{-1}$ (equivalently, closed under the operation and inverses).

Q7. (2 marks) Define a cyclic group with an example. **Solution — full explanation.**

Definition. A group G is **cyclic** if some element $a \in G$ generates it: $G = \langle a \rangle = \{a^n : n \in \mathbb{Z}\}$. The element a is a **generator**. Every cyclic group is abelian. **Example.** $(\mathbb{Z}_5, +_5)$ is cyclic, generated by 1: $\langle 1 \rangle = \{1, 2, 3, 4, 0\}$. **Final answer.** A cyclic group is generated by a single element (and is always abelian).

Q8. (2 marks) Define a coset. **Solution — full explanation.**

Definition. Let H be a subgroup of $(G, *)$ and $a \in G$. The **left coset** of H determined by a is $aH = \{a * h : h \in H\}$. The **right coset** is $Ha = \{h * a : h \in H\}$. **Key fact.** Two left cosets are either identical or disjoint, and $|aH| = |H|$.



Final answer. A coset is $aH = \{a * h : h \in H\}$; the cosets partition G .

Q9. (2 marks) State Lagrange's theorem. **Solution — full explanation.**

Theorem. If H is a subgroup of a finite group G , then $|H|$ **divides** $|G|$, and the number of distinct left cosets is the index $[G : H] = |G|/|H|$. **Reason.** The left cosets partition G into blocks each of size $|H|$, so $|G| = (\text{number of cosets}) \times |H|$. **Final answer.** $|H|$ divides $|G|$; index $[G : H] = |G|/|H|$.

Q10. (2 marks) Define a normal subgroup and a quotient group. **Solution — full explanation.**

Definitions.

- **Normal subgroup:** $H \leq G$ with $aH = Ha$ for every $a \in G$ (equivalently $aHa^{-1} \subseteq H$).
- **Quotient group:** if H is normal, the cosets $G/H = \{aH : a \in G\}$ form a group under $(aH)(bH) = (ab)H$, of order $|G|/|H|$.

Final answer. H normal $\Leftrightarrow aH = Ha$; then G/H is a group.

Q11. (2 marks) Define a homomorphism and an isomorphism. **Solution — full explanation.**

Definitions. For groups $(G, *)$ and $(G', \circ)$:

- **Homomorphism:** a map $\varphi : G \rightarrow G'$ with $\varphi(a * b) = \varphi(a) \circ \varphi(b)$ (preserves the operation).
- **Isomorphism:** a homomorphism that is also a **bijection**; then $G \cong G'$.

Final answer. A homomorphism preserves the operation; an isomorphism is a bijective homomorphism.

Q12. (2 marks) Define the Hamming distance and the minimum distance of a code. **Solution — full explanation.**

Definitions.

- **Hamming distance** $d(x, y)$: the number of positions in which the binary words x and y differ.
- **Minimum distance** of a code: the smallest Hamming distance between any two distinct codewords.

Final answer. Hamming distance = number of differing bits; minimum distance = smallest such distance in the code.

Q13. (2 marks) What is a group code? **Solution — full explanation.**

Definition. A code is a **group code** if the set of all its codewords forms a **group under bitwise XOR** $\oplus$ (closed under $\oplus$, with the zero word as identity). **Useful property.** In a group code the minimum distance equals the minimum weight of a non-zero codeword. **Final answer.** A group code is a set of codewords forming a group under XOR.

Q14. (2 marks) A code has minimum distance d . How many errors can it detect and correct? **Solution — full explanation.**

Result. A code with minimum distance d :

- can **detect** up to $d - 1$ errors;
- can **correct** up to $\left\lfloor \frac{d - 1}{2} \right\rfloor$ errors.

Final answer. Detects $d - 1$ errors; corrects $\lfloor (d - 1)/2 \rfloor$ errors.

Part B — Long Questions (7 Marks)

Q15. (7 marks) Prove that modulo 5 under addition $(\mathbb{Z}_5, +_5)$ forms a group. Further prove that the group is cyclic. **Solution — full explanation.**

What the question asks: verify the four group axioms, then exhibit a generator. **Group axioms.**

- **Closure:** every $a +_5 b$ lies in $\{0, 1, 2, 3, 4\}$ (check the Cayley table).
- **Associativity:** $(a +_5 b) +_5 c = a +_5 (b +_5 c)$, inherited from ordinary addition.
- **Identity:** 0, since $a +_5 0 = a$.
- **Inverse:** $0^{-1} = 0$, $1^{-1} = 4$, $2^{-1} = 3$, $3^{-1} = 2$, $4^{-1} = 1$ (each pair sums to $5 \equiv 0$).

All four hold, so $(\mathbb{Z}_5, +_5)$ is a group. **Cyclic.** The generator 1 produces the whole group: 1, 2, 3, 4, 0 (mod 5). Hence $\mathbb{Z}_5 = \langle 1 \rangle$. **Final answer.** $(\mathbb{Z}_5, +_5)$ is a cyclic (hence abelian) group generated by 1.

Q16. (7 marks) Show that $(G, \oplus_8)$ is an abelian group, where $G = \{0, 1, 2, 3, 4, 5, 6, 7\}$. **Solution — full explanation.**

What the question asks: verify that addition modulo 8 on $\{0, \dots, 7\}$ is an abelian group. **Axioms.**

- **Closure:** $a +_8 b \in \{0, \dots, 7\}$ always.
- **Associativity:** inherited from integer addition.
- **Identity:** 0.
- **Inverse:** $a^{-1} = 8 - a$ (with $0^{-1} = 0$), since $a + (8 - a) = 8 \equiv 0 \pmod{8}$.
- **Commutative:** $a +_8 b = b +_8 a$.

Final answer. $(G, \oplus_8) = (\mathbb{Z}_8, +_8)$ is an abelian group.

Q17. (7 marks) Show that the set of all positive rational numbers forms an abelian group under $a * b = \frac{ab}{2}$. **Solution — full explanation.**

What the question asks: verify that $(\mathbb{Q}^+, *)$ is an abelian group. **Axioms.**

- **Closure:** $ab/2$ is a positive rational for $a, b > 0$.
- **Associativity:** $(a * b) * c = \frac{abc}{4} = a * (b * c)$.
- **Identity:** $e = 2$, since $a * 2 = \frac{2a}{2} = a$.
- **Inverse:** $a^{-1} = \frac{4}{a}$, since $a * \frac{4}{a} = \frac{a \cdot 4/a}{2} = 2 = e$.
- **Commutative:** $a * b = \frac{ab}{2} = \frac{ba}{2} = b * a$.

Final answer. $(\mathbb{Q}^+, *)$ is an abelian group with identity 2 and inverse $a^{-1} = 4/a$.

Q18. (7 marks) Show that the set of integers does not form a group under multiplication. **Solution — full explanation.**

What the question asks: show $(\mathbb{Z}, \times)$ fails to be a group. **Check.** $(\mathbb{Z}, \times)$ is closed, associative, and has identity 1. But it fails the **inverse** axiom: for $2 \in \mathbb{Z}$ we would need an integer b with $2b = 1$; no such integer exists (since $\frac{1}{2} \notin \mathbb{Z}$). So not every element has an inverse. **Final answer.** $(\mathbb{Z}, \times)$ is not a group (elements like 2 have no inverse in $\mathbb{Z}$).

Q19. (7 marks) Prove that the fourth roots of unity $\{1, -1, i, -i\}$ form an abelian multiplicative group. **Solution — full explanation.**

What the question asks: verify the group axioms for $G = \{1, -1, i, -i\}$ under multiplication. **Axioms.**

- **Closure:** $i \cdot i = -1$, $i \cdot (-i) = 1$, $(-i)(-i) = -1$; all results lie in G (indeed $G = \{i^n\}$).
- **Associativity:** complex multiplication is associative.
- **Identity:** 1.
- **Inverse:** $1^{-1} = 1$, $(-1)^{-1} = -1$, $i^{-1} = -i$, $(-i)^{-1} = i$.
- **Commutative:** complex multiplication is commutative.

So G is an abelian group of order 4 (cyclic, generated by i). **Final answer.** $\{1, -1, i, -i\}$ is a cyclic abelian group under multiplication.

Q20. (7 marks) Check whether the set of all 2×2 matrices over $\mathbb{Q}$ under matrix multiplication forms a group. **Solution — full explanation.**

What the question asks: decide whether the full set of 2×2 rational matrices is a group under multiplication. **Check.** Closure, associativity and identity (I) all hold. But the **inverse** axiom fails: a matrix with determinant 0 (e.g. the zero matrix) has no multiplicative inverse. So not every element is invertible. **Final answer.** No — singular matrices have no inverse, so the inverse

axiom fails. (The *invertible* 2×2 matrices do form a group, $GL(2, \mathbb{Q})$.)

Q21. (7 marks) What is a group code? Given the set of codewords $\{000, 111\}$, check whether this set forms a group under bitwise XOR. **Solution — full explanation.**

Group code. A code whose codewords form a group under bitwise XOR $\oplus$ (closed under $\oplus$, with the zero word as identity). **Check closure** for $\{000, 111\}$:

$$000 \oplus 000 = 000, \quad 000 \oplus 111 = 111, \quad 111 \oplus 000 = 111, \quad 111 \oplus 111 = 000.$$

Every result is again a codeword and 000 is the identity. **Final answer.** Yes — $\{000, 111\}$ is a group code under XOR (minimum weight 3).

Q22. (7 marks) Show that the (2,5) encoding function defined by $e(00) = 00000$, $e(01) = 01110$, $e(10) = 10101$, $e(11) = 11011$ is a group code. **Solution — full explanation.**

What the question asks: show the codewords are closed under XOR. **Check every pair.**

$$01110 \oplus 10101 = 11011, \quad 01110 \oplus 11011 = 10101, \quad 10101 \oplus 11011 = 01110,$$

and $00000 \oplus c = c$, $c \oplus c = 00000$ for each codeword c . **Conclusion.** Every XOR of two codewords is again one of $\{00000, 01110, 10101, 11011\}$, with 00000 as identity. So the codewords form a group under XOR. **Minimum weight:** the lightest non-zero codeword has weight 3, so $d = 3$ (detects 2 errors, corrects 1). **Final answer.** Yes — it is a group code with minimum distance $d = 3$.

Q23. (7 marks) For the (2,4) encoding function $e(00) = 0000$, $e(10) = 0110$, $e(01) = 1011$, $e(11) = 1100$, determine how many errors can be detected and corrected. **Solution — full explanation.**

What the question asks: find the error detection/correction capability. **Step 1 — minimum distance.** Weights of the non-zero codewords: $w(0110) = 2$, $w(1011) = 3$, $w(1100) = 2$. In a group code the minimum distance equals the minimum weight, so $d = 2$. **Step 2 — capabilities.**

- Detects up to $d - 1 = 1$ error.
- Corrects up to $\left\lfloor \frac{d-1}{2} \right\rfloor = 0$ errors.

Final answer. Detects 1 error; corrects 0 errors (since $d = 2$).

Q24. (7 marks) Construct a binary linear code using a generator matrix, and explain how syndrome decoding can correct errors. **Solution — full explanation.**

What the question asks: build a linear code and describe syndrome decoding. **Generator matrix** (a (2,5) code):

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

A message x is encoded as $c = xG$: $00 \rightarrow 00000$, $10 \rightarrow 10110$, $01 \rightarrow 01011$, $11 \rightarrow 11101$. Each non-zero codeword has weight 3, so $d = 3$. **Parity-check matrix** with $GH^T = 0$:

$$H = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

Syndrome decoding. For a received word r , compute the syndrome $s = Hr^T$.

- If $s = 0$: no error detected.
- Otherwise s equals the column of H at the error position; flip that bit.

Final answer. Encode by $c = xG$; the syndrome $s = Hr^T$ locates a single error so it can be corrected.

Q25. (7 marks) State Lagrange's theorem. If G is a group of order 10 and H a subgroup of order 5, how many distinct left cosets does H have? **Solution — full explanation.**

Lagrange's theorem. If H is a subgroup of a finite group G , then $|H|$ divides $|G|$, and the number of distinct left cosets is $[G : H] = |G|/|H|$. **Apply.** Here $|G| = 10$ and $|H| = 5$, so

$$[G : H] = \frac{|G|}{|H|} = \frac{10}{5} = 2.$$

Final answer. H has 2 distinct left cosets in G .